

KAISER

Engineering Journal

11/24/23

Engineering log

Stage of Team Development: Forming

PROBLEM: Traffic stop situations with police officers can be dangerous and nerve-wracking for both the officer & and the driver. Situations can become hostile, tense, and life-threatening depending on the circumstances.

RESEARCH: In 2021, 129 officers died in the line of duty, which is an increase of 29% from the previous year.

- 73 of those deaths were felonious, mostly firearms
- Felonious killings were the top of death for officers on duty, reaching a 25-year high in 2021.
- In 2021, 56 officers were accidentally killed. Most of these deaths resulted from motor vehicle crashes.
- The South (US) had 70 officer deaths, which is more than twice the rate of any other region.
- Breakdown of 2020 law enforcement shooting statistics:
 - 18 were investigating a suspicious person or responding to a domestic disturbance
 - 12 were shot while attempting an arrest, were ambushed, or shot in tactical situations
 - 4 shots responding to a robbery or burglary
 - 3 killed during traffic stops
 - 11 serving warrants, shot inadvertently, or shot responding to a mental health call

We decided to create a robot that helps diffuse any stress and anxiety that goes both ways for the officer and the person getting stopped. The robot would help perform minor traffic stops in place of the officer to prevent unnecessary confrontations. This would not replace traffic stops for important incidents like speeding or DUIs requiring confrontation and breathalyzers. The plan is to provide the robot with a screen, camera, microphone, and speaker to approach the driver and ask questions. It can scan cards and print citations while the officer can view the camera and control the robot from his car.

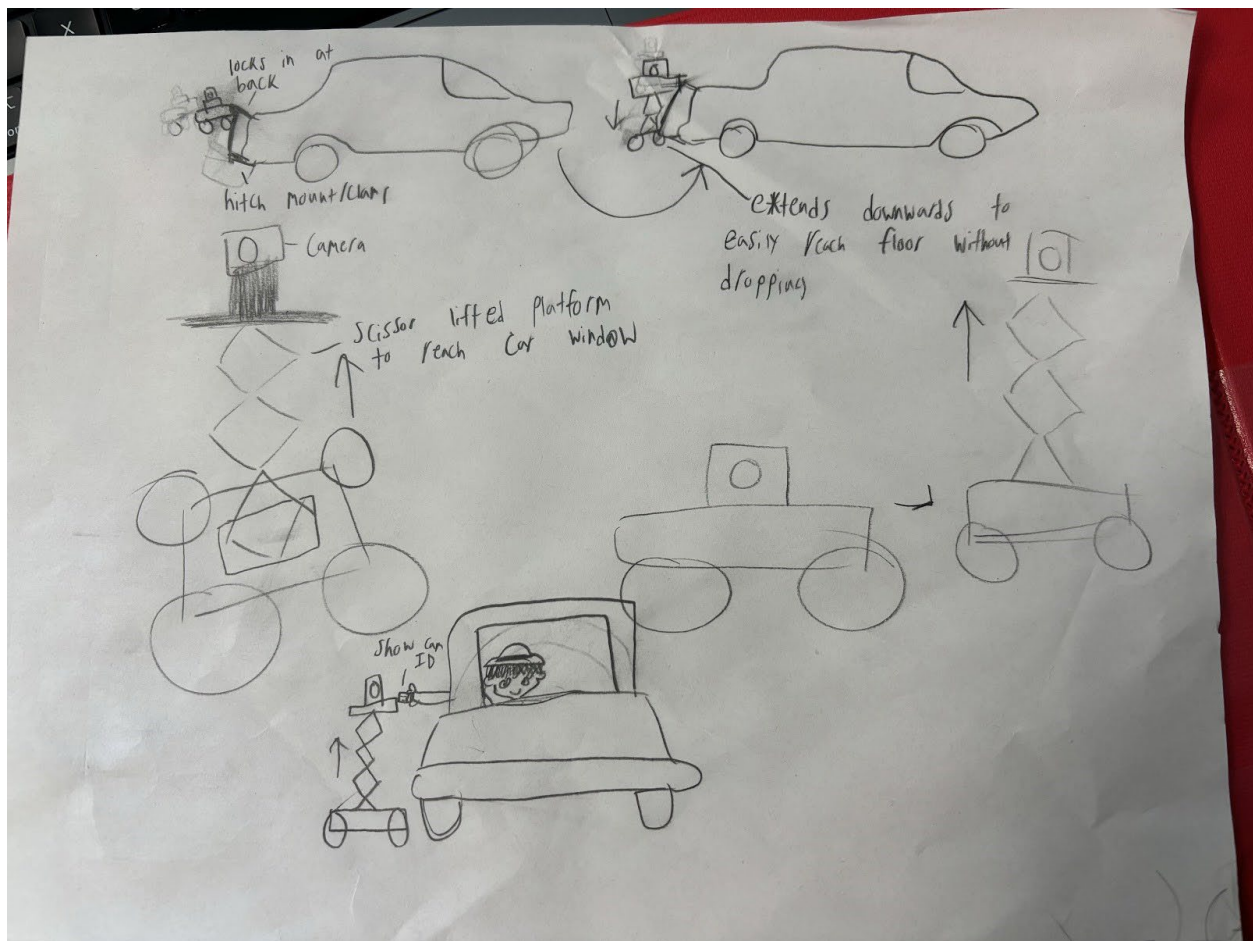


Figure 1. Our design sketches

Melanie Guerrero

11/25/2023

Engineering log

Stage of Team Development: Performing

Today, we had the privilege of talking to our school police officer about our idea. We presented the idea to an officer and asked him questions that helped us with our brainstorming phase.

RESEARCH:

<ADD TRANSCRIPT WITH OFFICER>

"I'm nervous when I get pulled over"

How often do you deal with traffic stops?

Before: average a day 7-8

What are things you consider when you conduct a traffic stop?

Unexplained Movement,

What are some things you look at when looking at a driver?

How do you genuinely feel when conducting a traffic stop?

Fear of the unknown, must be ready for the unknown

Ticket printer, camera, speaker, ID Scanner

Would want to include a view of the A pillar

Language barrier/ have code detect multiple languages

Hard of hearing/ screen

Broken door/ knocked down robot/ how much time it would take for the robot to get back up

False sense of security

How fast do officers attack during traffic stops?

Video of the robot working showing the robot working in motion

Melanie Guerrero

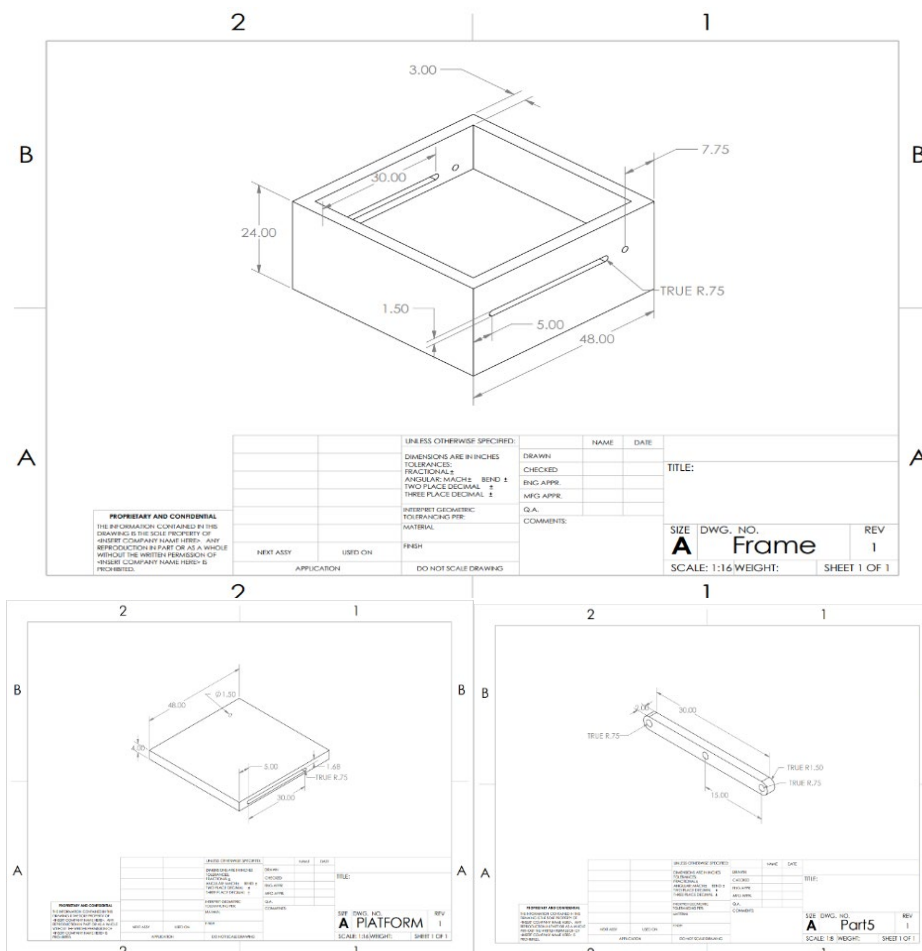
1/10/2024

ENGINEERING JOURNAL

STATE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: We need the correct measurements so our robot can be used to reach the height of a vehicle window while also being as compact as possible.

SOLUTION: We drafted a general shape and structure that we want our robot to use. The core feature is the scissor lift mechanism. We decided this was a good general size because it would allow our robot room for wheels, motors, and other electrical components without being too cramped. It also allows room for the arms to be 30" which is enough to make our robot reach a height of around 5' which is enough to reach the window of most vehicles. This will just be the general shape of our robot, so that instead of recreating it entirely we can simply make small alterations over time.



Figures 1, 2, and 3 show the technical drawing of our assembly

Coby Nguyen

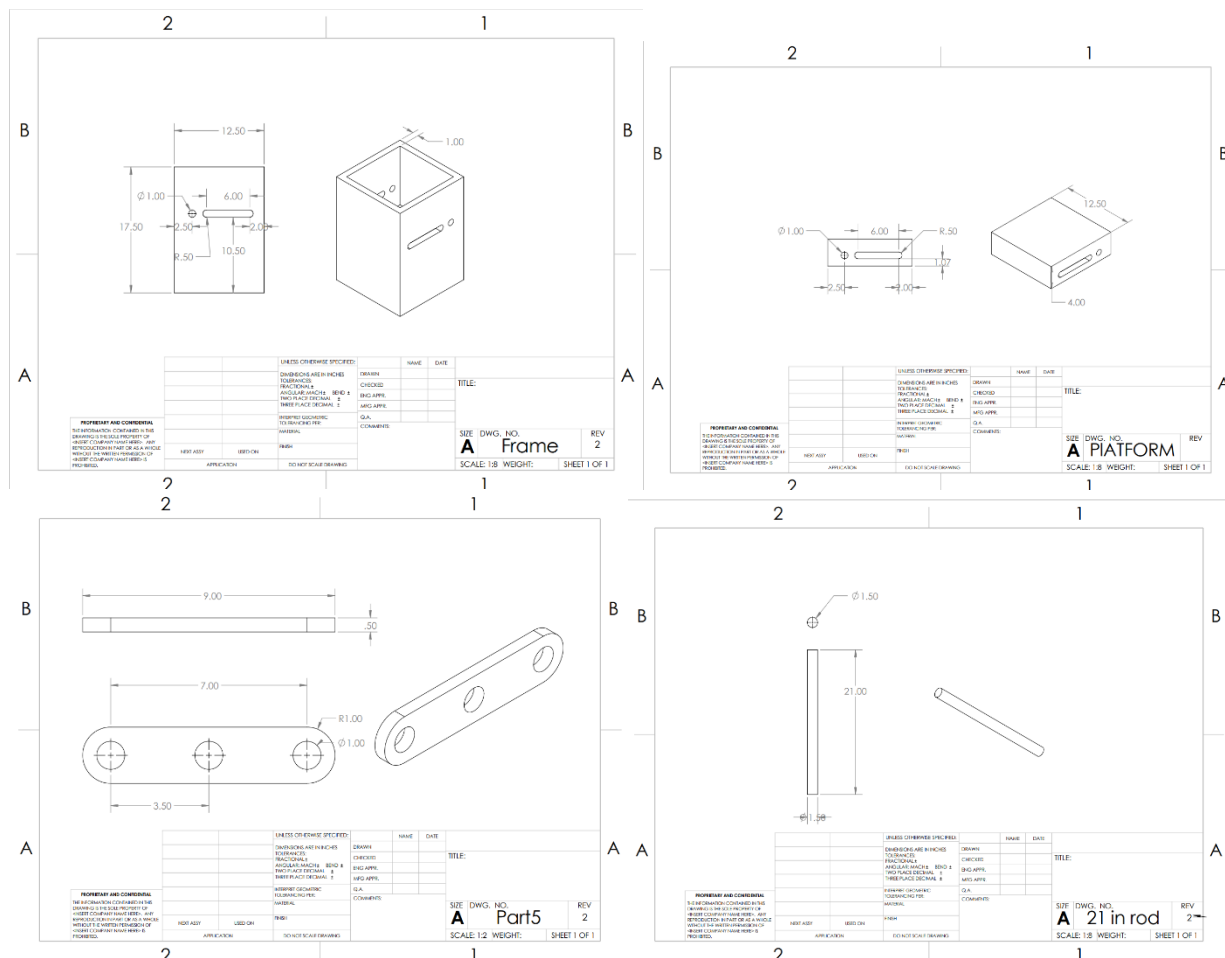
1/11/2024

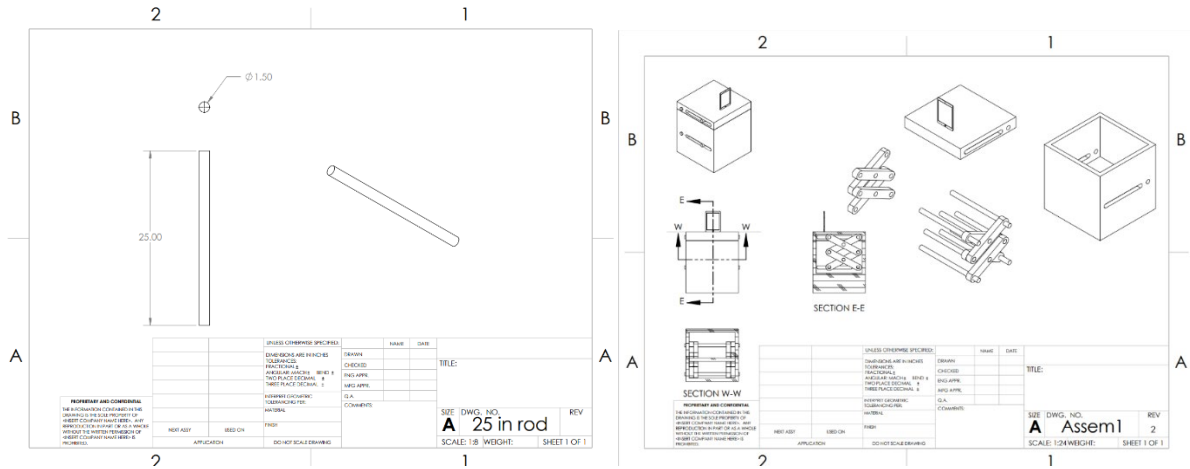
ENGINEERING JOURNAL

STATE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: The initial design for our robot was not the optimal size to fit the needs of our robot. Our robot needs to be compact, but large enough to fit the components of a scissor lift so it can extend outwards or back inwards. It must be small enough to sit in the back of a police car while just large enough to fit the iPad on the front.

SOLUTION: We increased the height of our frame so that it has more room for components such as wires and the scissor lift arm. We scaled down the length and width to 12.5" x 12.5" to also fit an iPad without being larger than it. Our frame is now 17.5" x 12.5" x 17.5". The arms are a total length of 9" which allows our assemble scissor lift to reach the height of a car window.





Figures 1-6. The technical drawing of our assembly



Luis Alvarez

1/16/2024

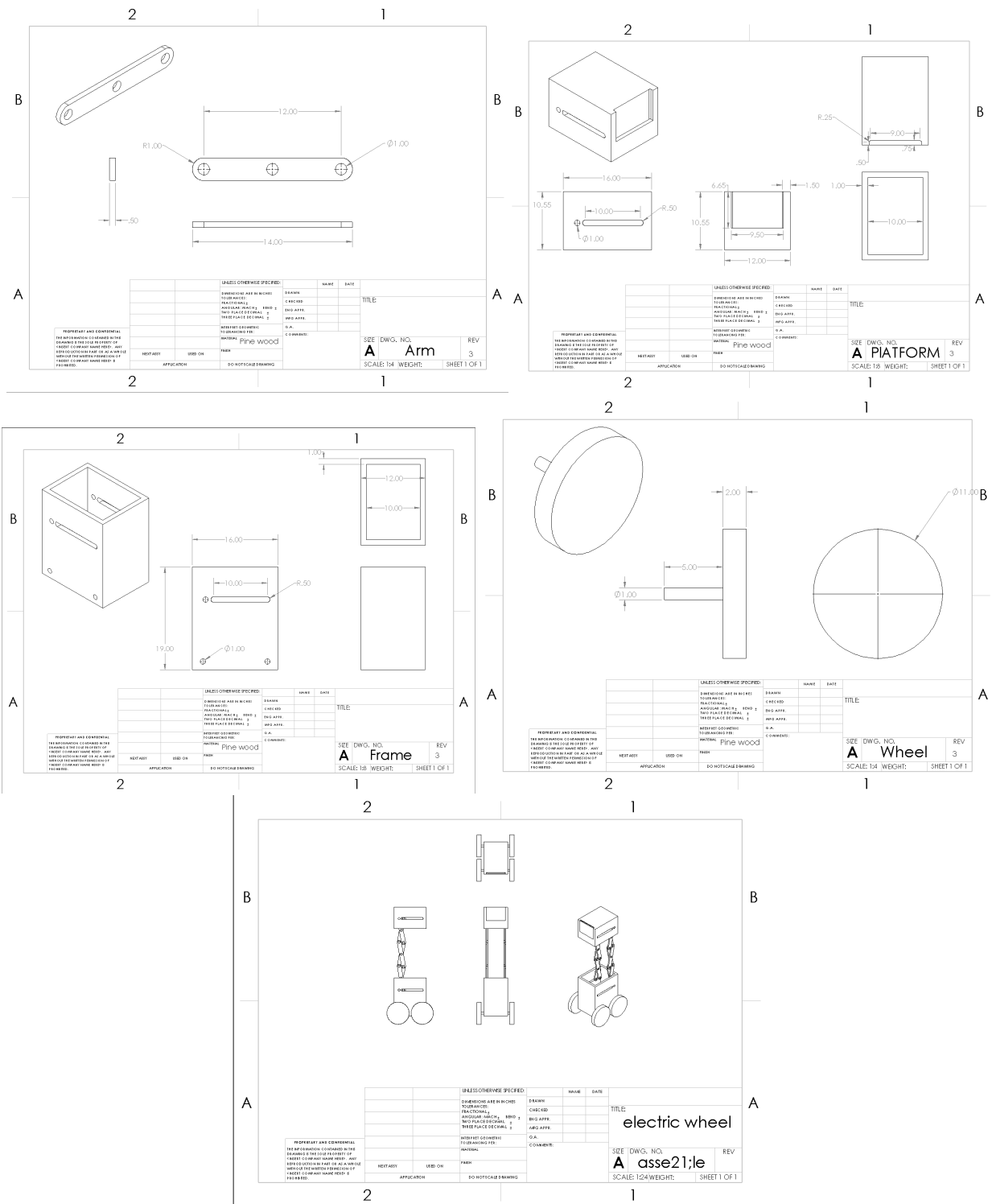
ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

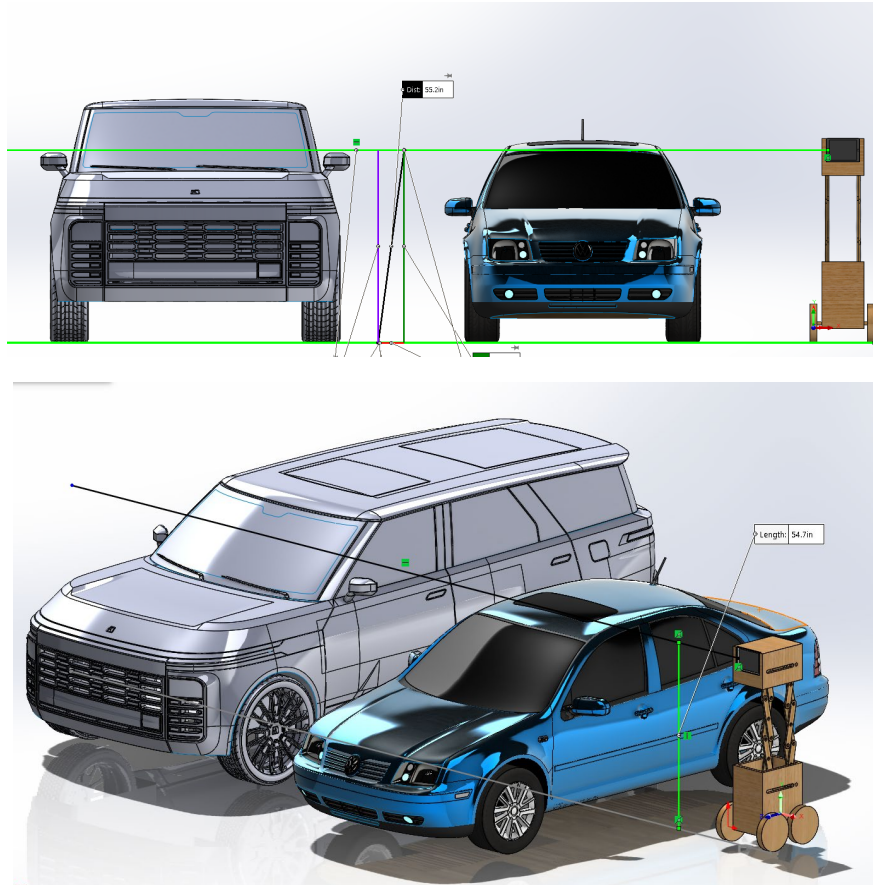
PROBLEM: The maximum height of our robot needs to account for larger cars which is not possible with the size of our frame. The size of our frame affects how large our arms can be, and the number of arms we can add, meaning to increase our height we need to increase the size of our components

RESEARCH: We investigated the sizes of several cars. We found online scale models of a Lincoln Aviator and Volkswagen Jetta which show us a general range of car sizes. This means we can use these heights as a minimum and maximum height for our robot.

SOLUTION: The frame of our robot is now longer to allow more arms of greater length. The We increased the height of the frame and the length of the arms. The height of the frame is now 19" and the length of the arms is 12". We found that the largest diameter our wheels can be is 12" without them touching. Depending on where we place the wheel, this gives us a bit more height too. We likely will not be making wheels with wood though. These pictures show the height that our camera reaches, and the line shows where the camera reaches compared to the windows. With these measurements, the camera will be able to reach 55" which is the perfect height.



Figures 1-5. The technical drawing of our assembly



Figures 5 and 6. The height of our camera compared to the Lincoln Aviator and Volkswagen Jetta

Coby Nguyen

1/22/2024

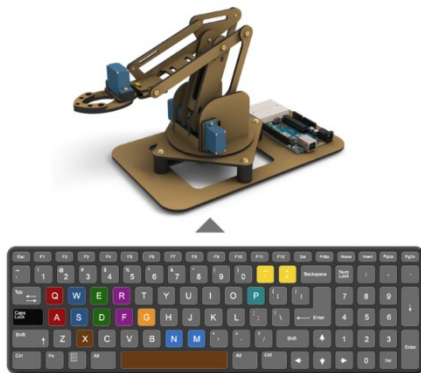
ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: We kept wondering about the ways we would control our robot. At first, we thought it would be easier to code a software program that the police officer can easily download. However, how would we create this software program while also controlling the robots through the computer's keys? How would we start this entire process?

RESEARCH: I found an article that controlled the robot through the computer's key which is what we wanted! We believed that C++ would be the most suitable language to use to code this entire software. However, in our situation, we want to build a robot that can be controlled through software without any wires connected to our robot. In the picture below, you can see that the Arduino is attached to the robot and would then have to be attached to the computer by a USB cable.

SOLUTION: We have decided that it would be a better idea to prioritize building the robot first instead of focusing on building software. Our teacher suggested we use a controller just so we can easily visualize each movement of the robot with a controller.



Melanie Guerrero

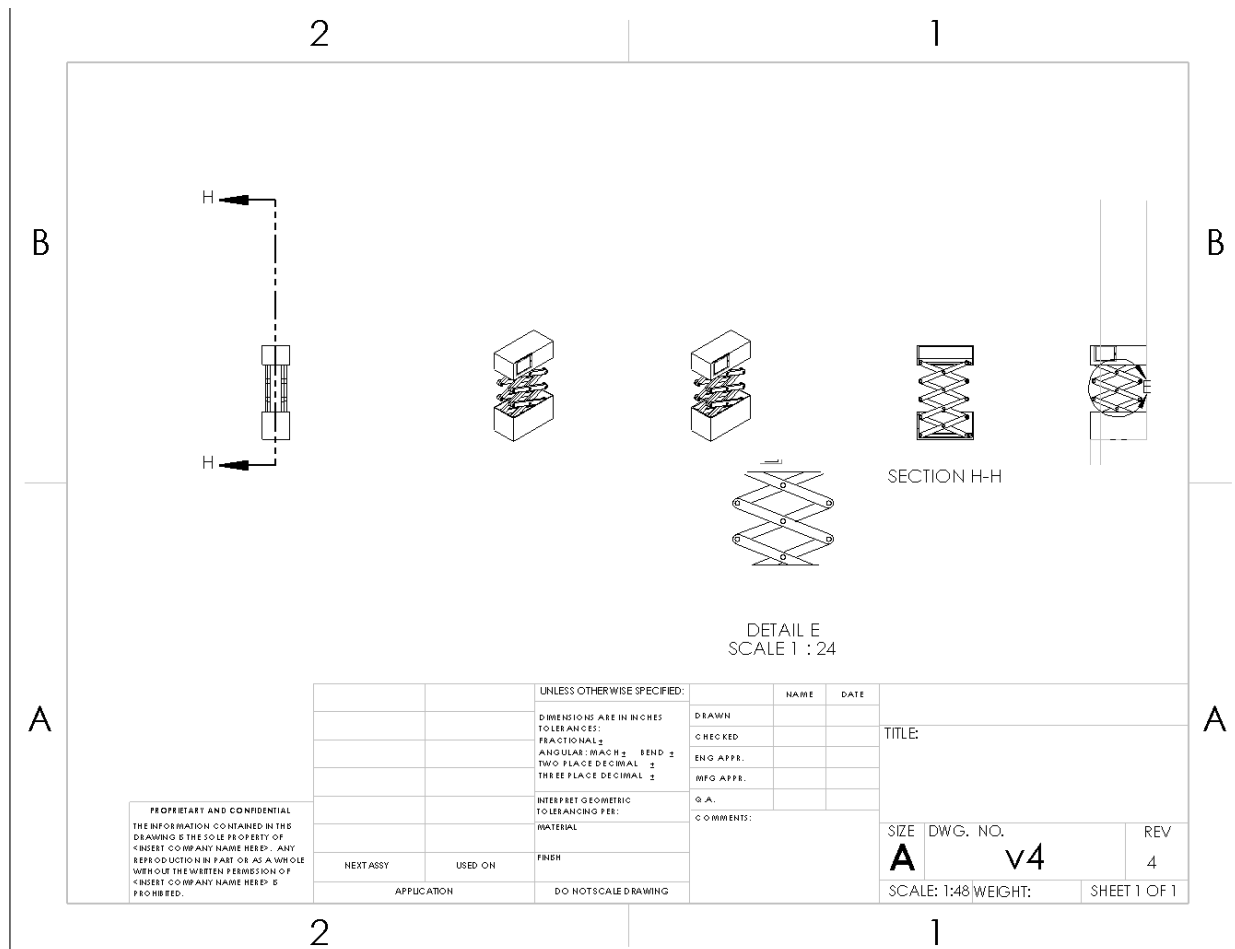
1/24/2024

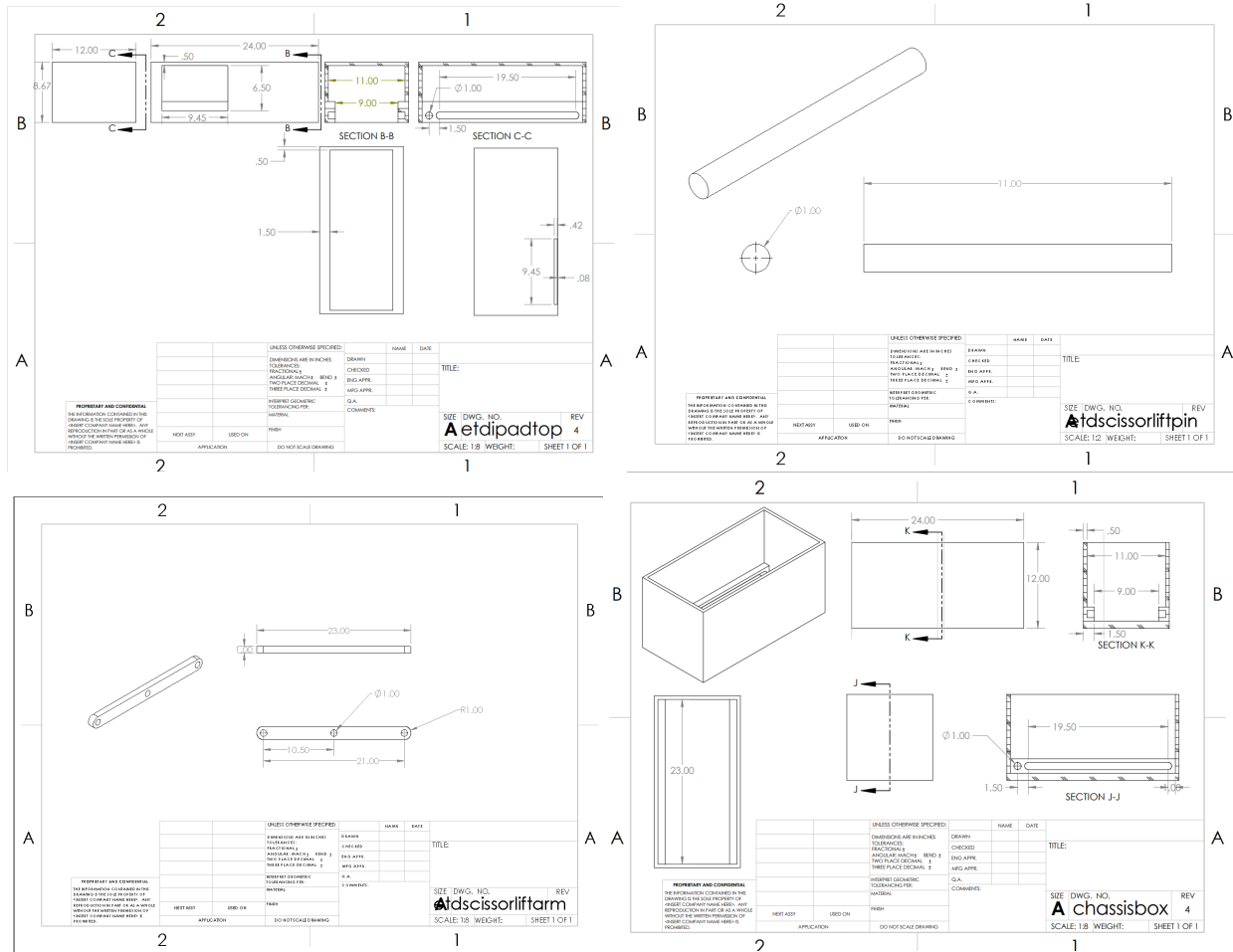
ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

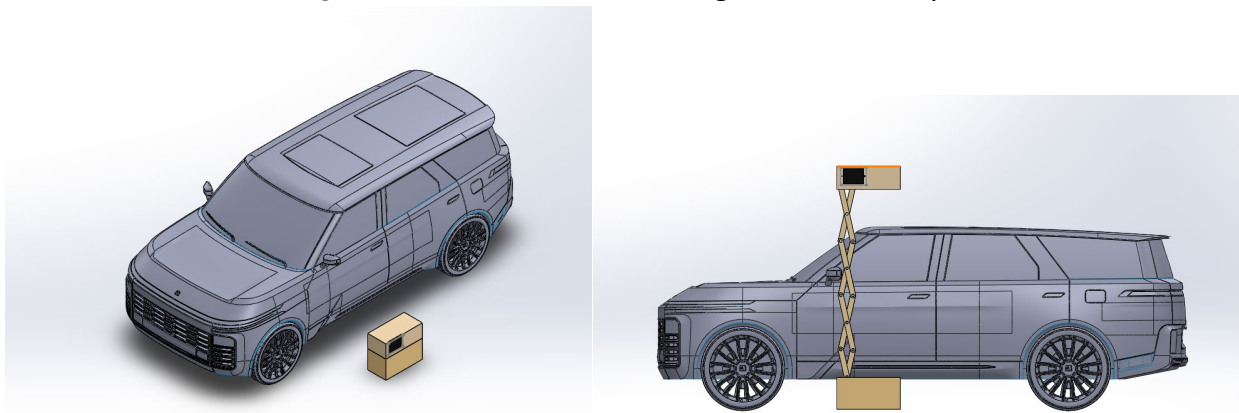
PROBLEM: Our robot reaches above the average car. This means there is room to make the robot compact and use the minimum amount of space possible while reaching the car window.

SOLUTION: Making this smaller and decreasing the number of arms should allow our robot to find a balance between height and compactness. We increased the size of the robot from 19" to 2' while the width remained 1'. The reason we increased the size of the robot is to allow room for longer arms. Longer arms mean that the more the anchor slot moves, the greater the robot extends. This means our robot can extend faster and higher than previously. This allows for the robot to reach a maximum height of 90" which is more than enough to reach a large majority of vehicles.





Figures 1-5. The technical drawing of our assembly.



Figures 6 and 7. The height of our camera compared to the Lincoln Aviator

Luis Alvarez

2/2/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: Using a hydraulic system for our scissor lift is difficult for these reasons:

- Complicated to learn the inner processes
- Requires large space to store the tank
- Capable of leaking
- Expensive to purchase
- Difficult to implement with our knowledge

RESEARCH: There are other potential ways to make our scissor lift move. It may be possible to simply use an electric motor like how electric scissor lifts work. One suggestion our teacher came up with is the usage of a linear actuator. Linear actuators move in a straight line to push and pull things. This is useful because our scissor arms are anchored at the edges, which only move in a line. We can use these linear actuators to push and pull the scissor lift, which would then cause it to move up and down. This achieves the same function we planned to do with the hydraulic motors but is less complicated to understand and implement.

Our robot is 24" wide, but the slot for the scissor arms and rod to slide in is 19.5". This means we need an actuator length of less than that.

Model link: <https://grabcad.com/library/linear-actuator-33>

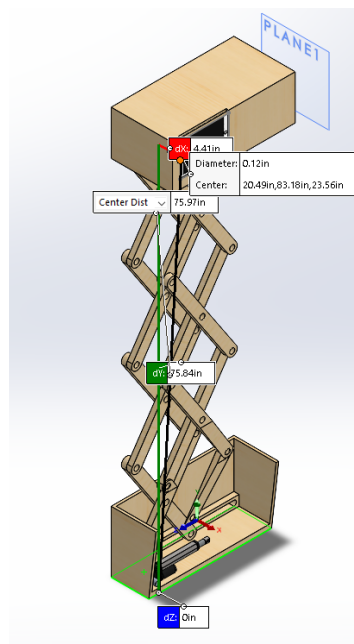


Figure 1. The height of the robot when the actuator is fully retracted

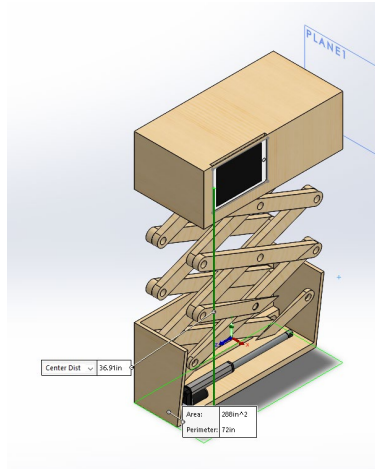


Figure 2. The height of the robot when the actuator is fully extended.

The closed length of this actuator is around 11" and the extended is around 19", making the stroke 250mm. This would allow the robot camera to reach a maximum height of around 76". When the linear actuator is fully extended, it retracts to only 36", meaning with this length the robot cannot become fully compact.

Another aspect of linear actuators to consider is the speed of the actual motors. Looking at a random selection of 12" actuators, they usually are around .40in/sec. This is very important because we want our robot to extend and contract as fast as possible so that the driver is not able to simply drive away and so that the police car can leave as soon as possible without wasting time.

SOLUTION: We will use a linear actuator rather than a hydraulic motor to lift our scissor lift. We are going to move the scissor lift through the bottom using the actuator, allowing our robot to close and open. We will need a 13' linear actuator with a stroke of 8' to be able to extend and retract to our minimum and maximum height specifications for the car. The engineering lab has a spare linear actuator we can use which saves us the effort in buying one because they are expensive. The retracted length of the actuator is 18' while the extended is 36', making the stroke 18'.

Coby Nguyen

2/5/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: Our engineering lab happened to have a real-life linear actuator in storage, but it is too large to let our robot fully extend. We also need to change where it pivots to push the robot. Even then, the current linear actuator does not fit.



Figure 1. The linear actuator we found in storage.

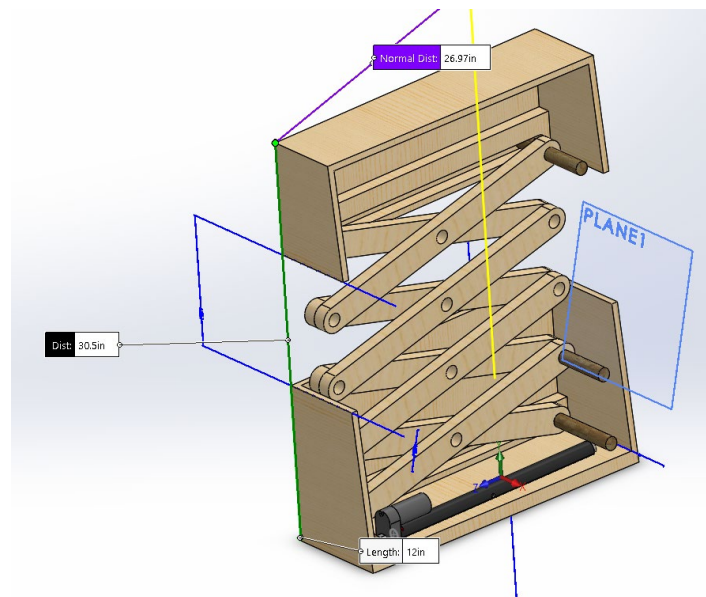


Figure 2. How the linear actuator can go when perfectly horizontal. Because the retracted length is 18', the highest our robot can go is 30.5' which is lower than our minimum.

RESEARCH: It is possible to connect our linear actuator diagonally rather than horizontally. We have space to be compact and fully extended. Other scissor lifts, mainly hydraulic ones, use this mechanism to lift their robot. Since the main motion of a linear piston is the same for both the hydraulic cylinder and the linear actuator, it is possible to implement this mechanism into our design.

Using the triangle anchor, we can attach that to one of the rod connectors of the scissor arms and have the actuator push or pull the triangle at an angle that will allow the scissor lift to move.

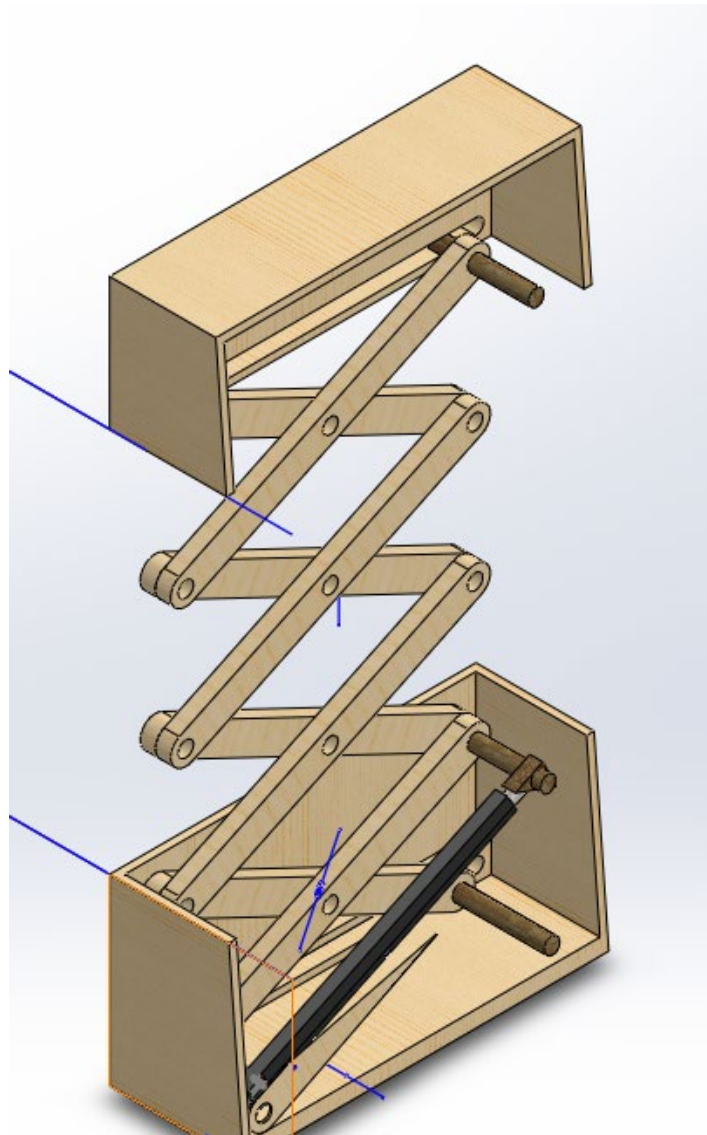


Figure 3. The linear actuator is connected to the pivots (ends) diagonally. We use the triangle wedges attached to the ends of the actuator as an anchor point for our actuator to push against.

However, our current linear actuator still does not fit within the constraints of our box. We also need to switch the side that the actuator is on. This is because the scissor lift leans to the left, so having the actuator push left from the right allows it to move the arms upwards at an angle.

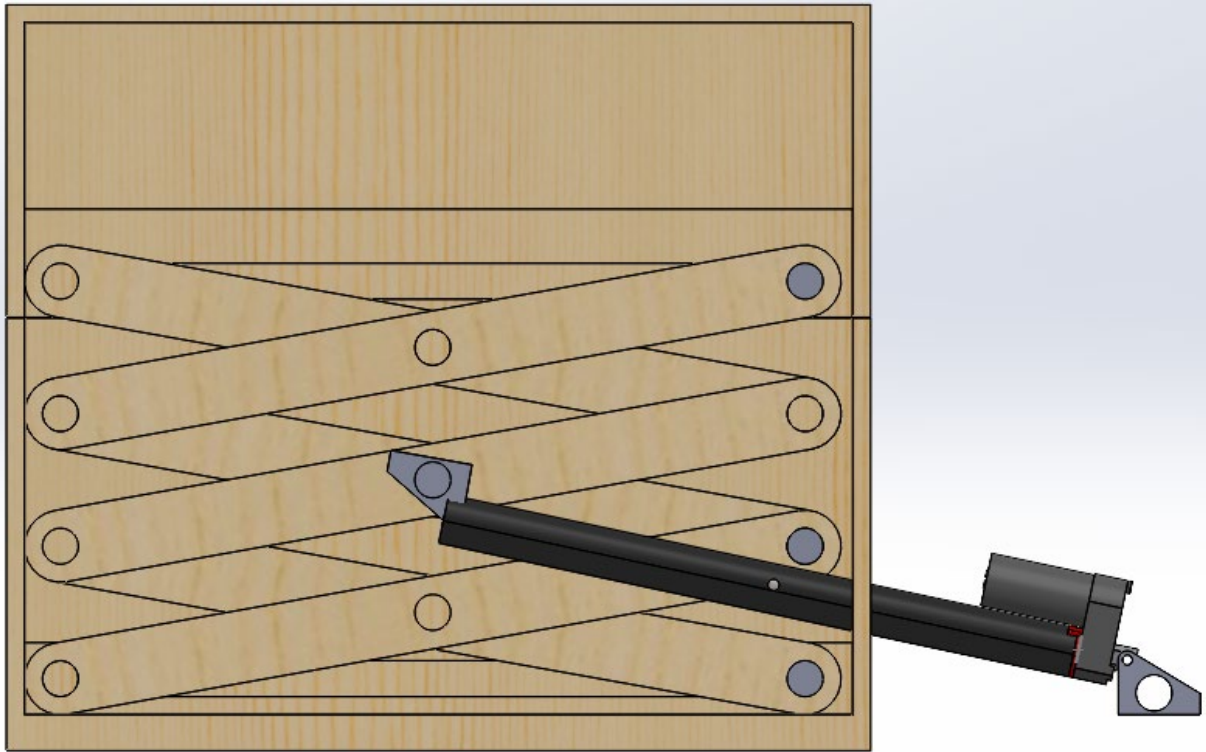


Figure 4. The minimum distance the triangle can be from the middle pivot to allow the robot to fully retract

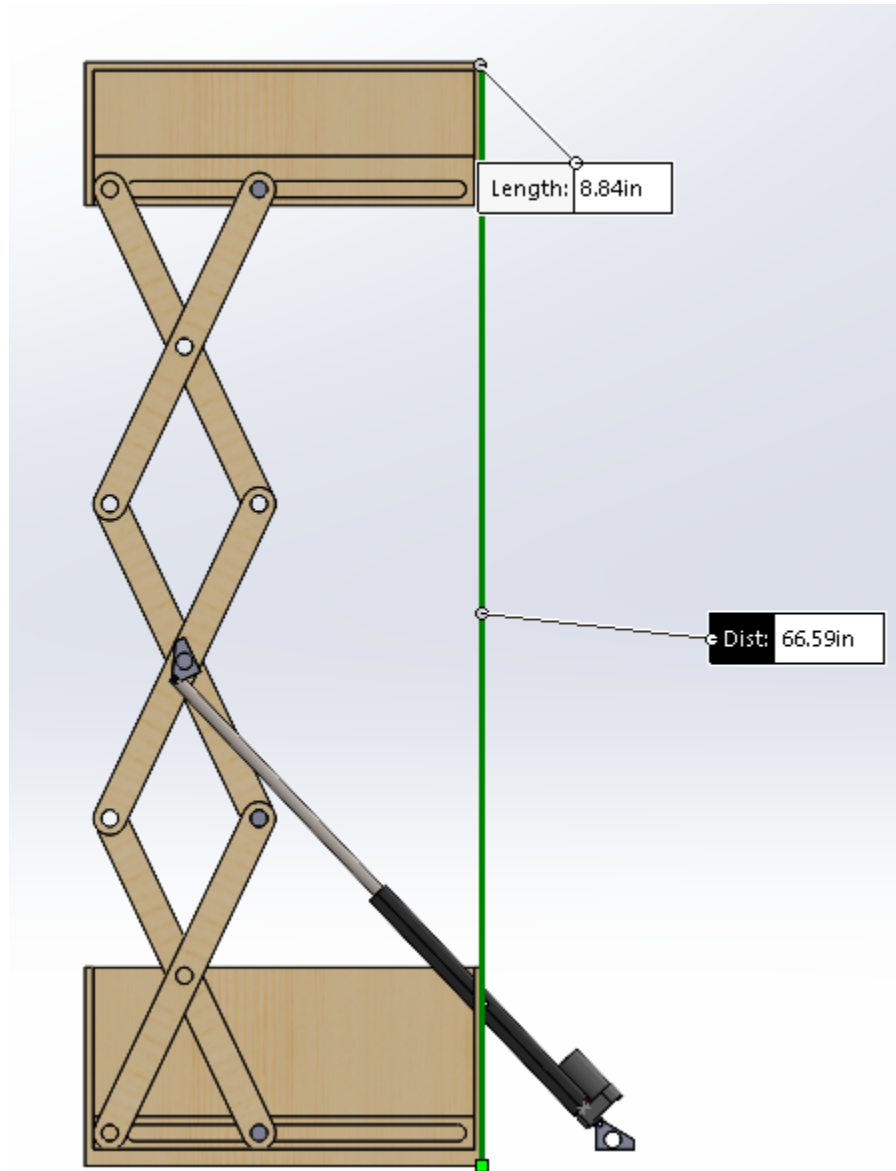


Figure 5. The maximum distance the actuator and robot can extend while keeping the distance to also fully compact.

We need to look for the perfect size for our linear actuator to fit in the box while still being able to extend fully. We found that the perfect size for our actuator is 18' retracted with a 14' stroke. Of course, we cannot find a linear actuator that fits the perfect dimensions, so we need to look at all the ones we can find that are close to these proportions and test the measurements in SolidWorks.

After testing several products on Amazon, we found that this "ECO-WORTHY" 14" linear actuator fit our proportions perfectly.



Figure 1.

https://www.amazon.com/gp/product/B00NM8H67C/ref=sw_img_1?smid=A1P855JR5KPOWZ&psc=1

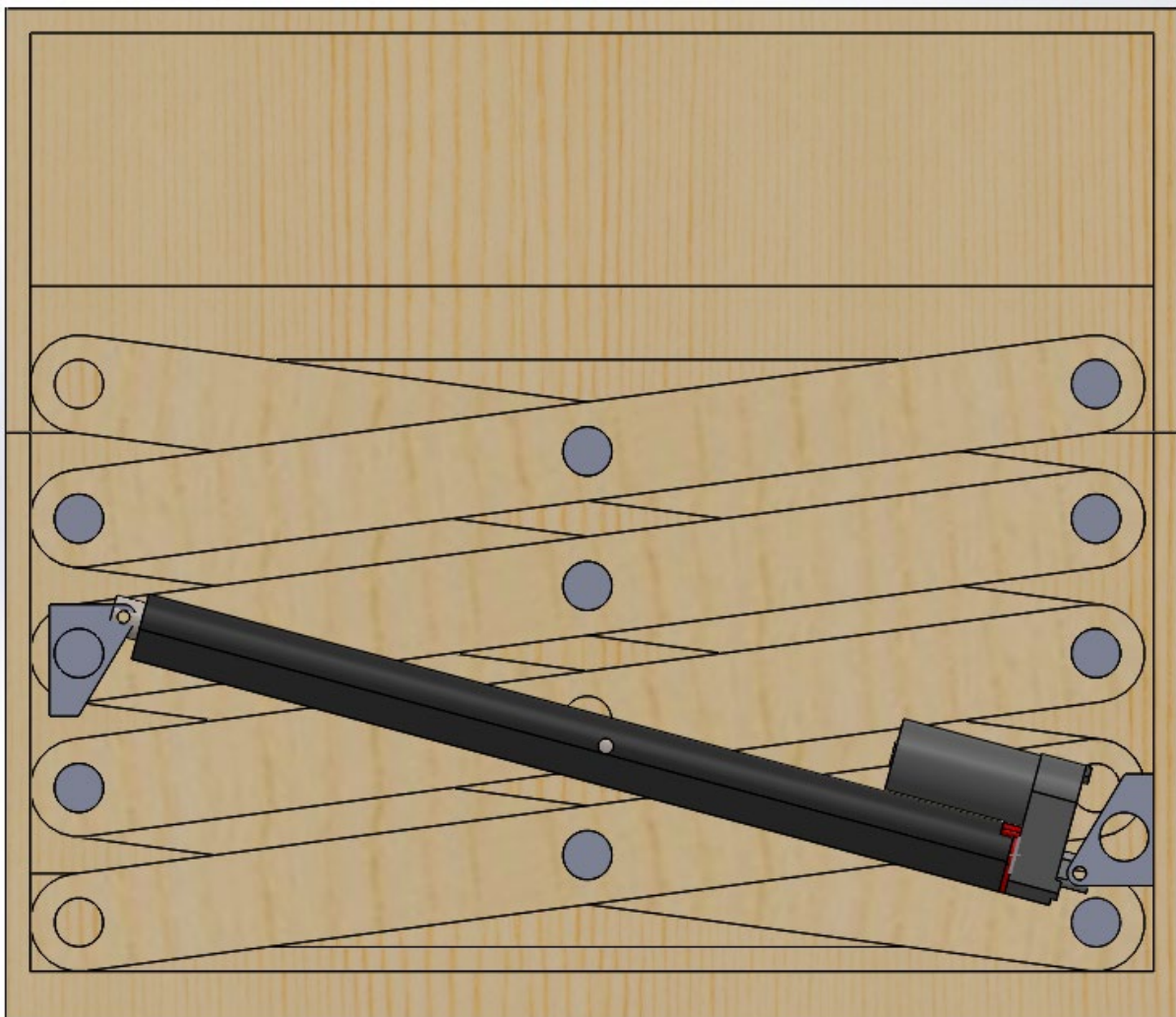


Figure 2. The listed linear actuator when it is fully retracted.

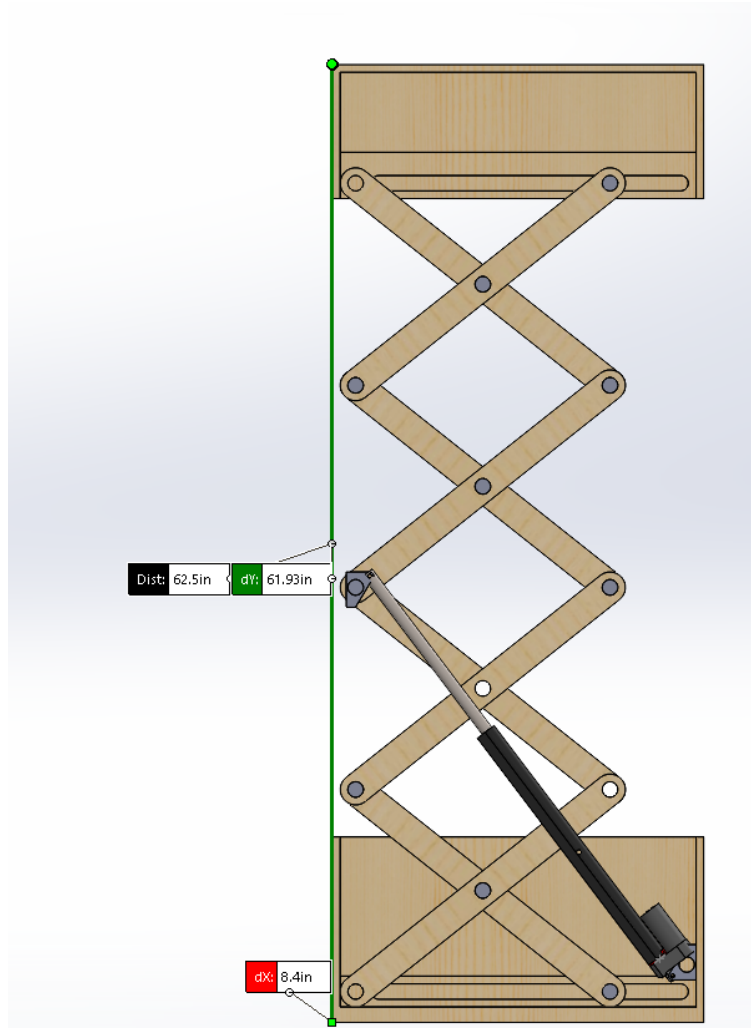


Figure 3. The listed linear actuator when it is fully extended. It can reach 5'2" because of the 14" stroke.

Luis Alvarez

2/7/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

One of our biggest limitations for this robot would be the unpredictable human behavior. People can exhibit unpredictable behavior during traffic stops, and a robot may struggle to interpret and respond appropriately to nuanced human emotions and intentions. However, we found a gas sensor that has a high sensitivity to LPG, Propane and Hydrogen, also could be used to Methane and other combustible gases. Being able to implement this gas sensor onto our robot will help solve one of our problems that we thought would be impossible.



Melanie

2/7/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: RESEARCH

PROBLEM: Becau

se of the size of our robot, it is not possible to have an actuator that has a short enough retraction length to pivot at a center point while also being able to push enough to extend our scissor lift to 4'5'. Even with the option of buying linear actuators, it is not possible because of our size constraints.

RESEARCH:

SOLUTION: We went back to our old design of having the linear actuator push against the scissor arms horizontally. The original reason we abandoned this design was because we wanted to use the linear actuator we had in storage, which was too large for this design. Now that we decided it was necessary and possible to purchase a new actuator we can go back to our original design.

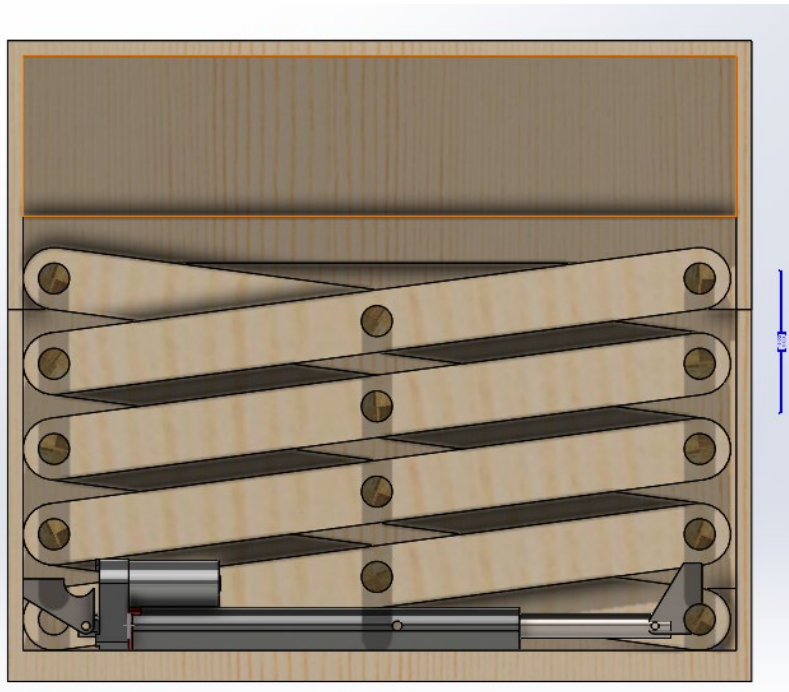


Figure 1. The robot when the linear actuator is horizontal and extended until the scissor lift is compact

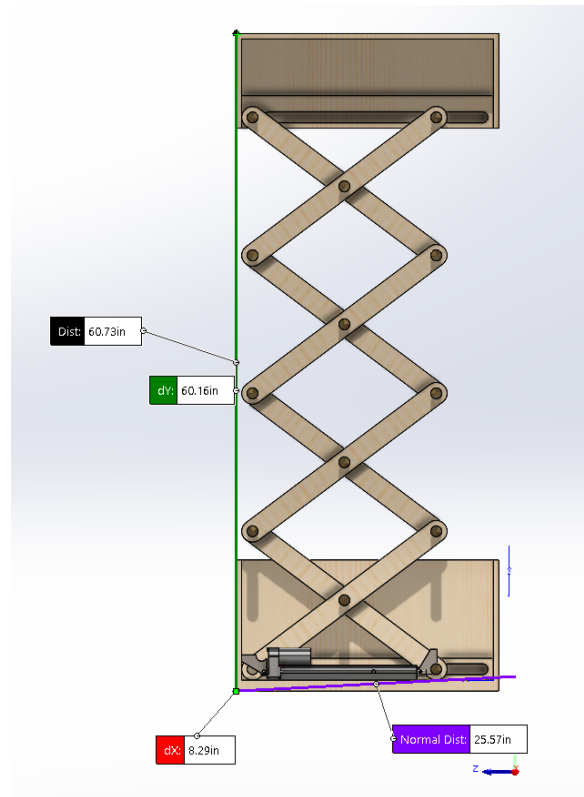
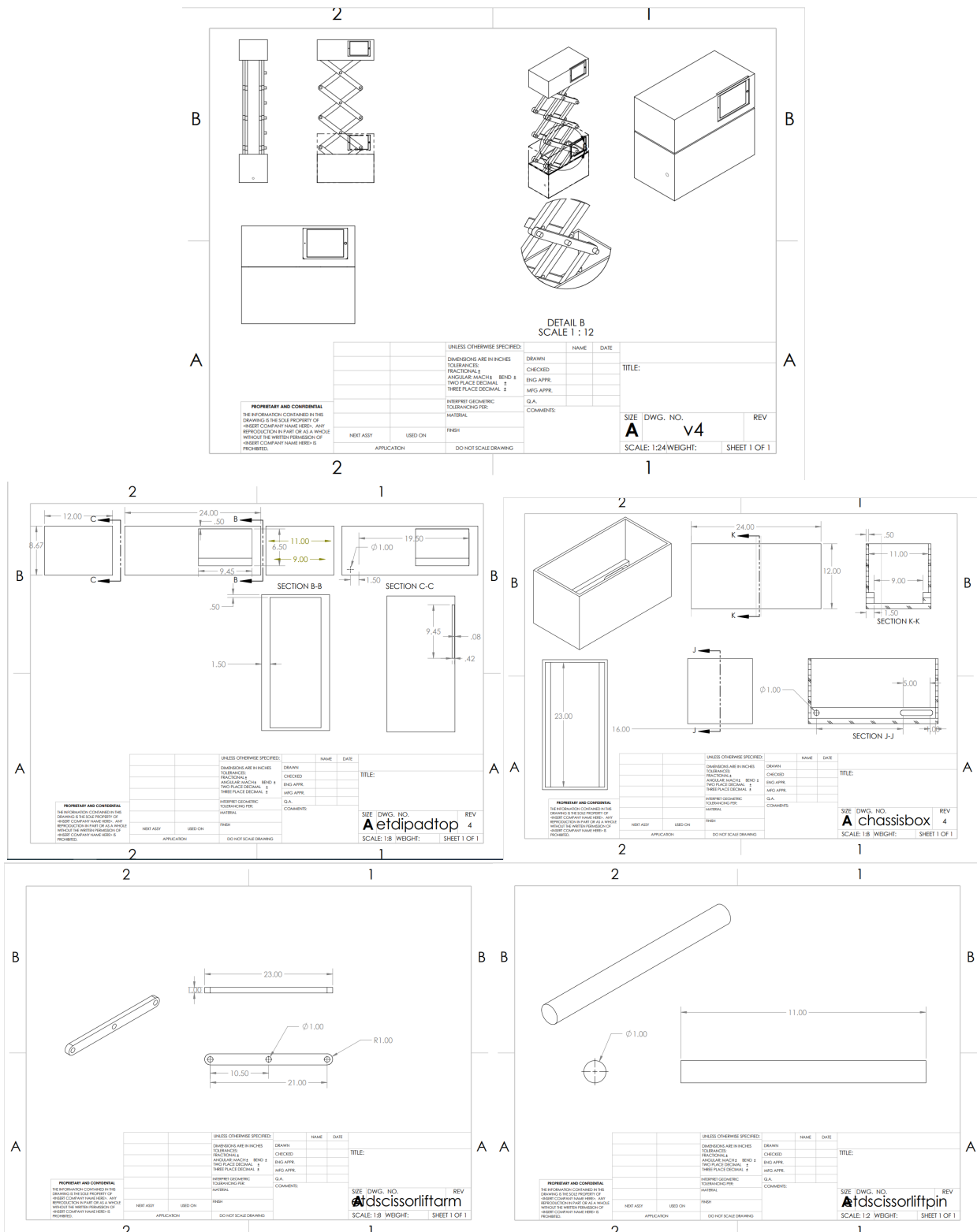


Figure 2. The scissor lift then the linear actuator is fully retracted.



Figures 5-7. The final technical drawings of our robot assembly

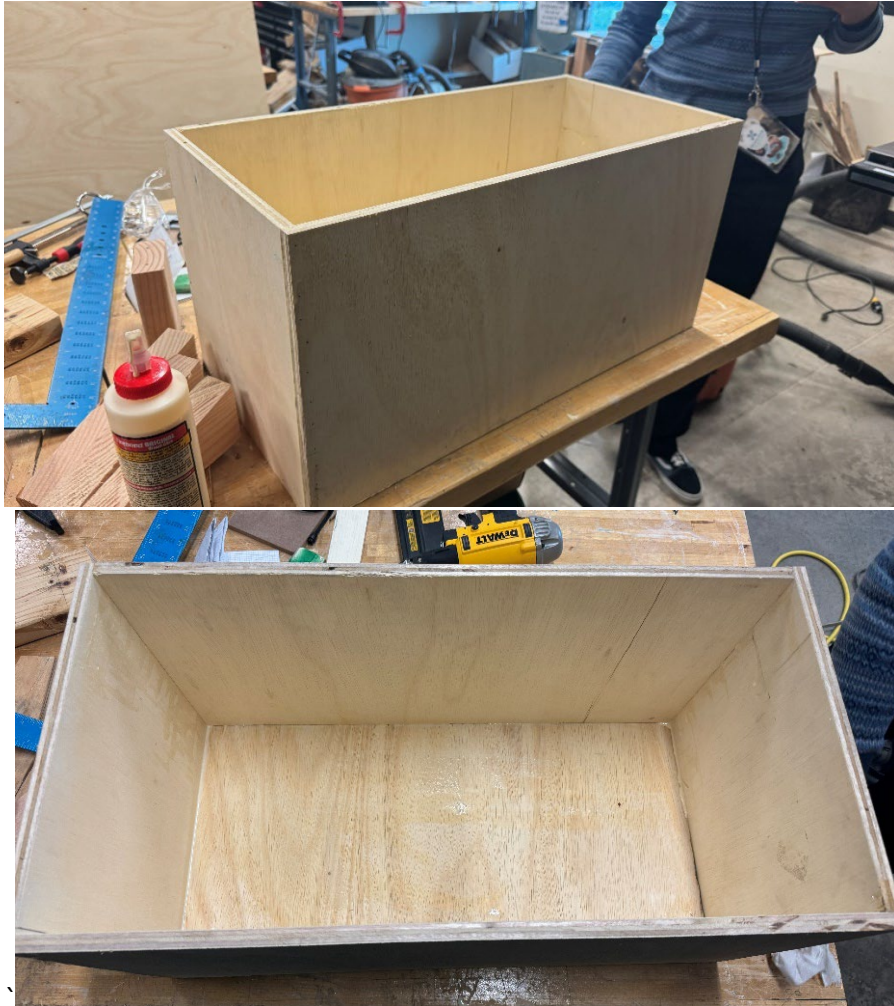
Coby Nguyen

2/7/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

We cut the bottom chassis using the saw and nailed the parts together. We used pine wood as our material, and we cut the dimensions listed on our technical drawings. The box is 23" x 12" x 12"



Figures 1 and 2. The end product of our bottom chassis



Figure 3. Luis cutting a part of the chassis with the saw.

Luis Alvarez

2/10/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

We continued cutting the pieces for our robot. We needed 16 arm pieces for the scissor lift, so we started by cutting those pieces out. We used the saw machine to cut off rectangles 2" wide and 23" long. We sanded them to be round on the edges which gives them room to turn.



Figures 1 and 2. Cutting and sanding the scissor arms out of wood

Luis Alvarez

2/14/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

We used the CNC machine to drill our separate slots that the bottom dowels would slide into. The reason we used a CNC machine rather than a drill is that we need our holes to be precise on the bottom to ensure the lift can go as high as possible. The holes also need to be perfect size and creating a straight slot is difficult without this machine.

We investigated how we would use the wheels and find a suitable motor for the wheels. The motors needed for the wheels are



Figures 1 and 2. The CNC machine cutting the slots into the wood

Melanie Guerrero

2/15/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

We drilled holes into our arms. Because the CNC machine already had cut a 1" hole in the bottom, we had to drill four of our arms with 1". The rest of the arms use $\frac{1}{2}$ " holes because the holes do not actually need to be that large and are still strong enough to hold.

We also drilled out a slot for the iPad screen to stick out of. Using the saw, we were able to cut a shape for just the screen, camera, and home button.

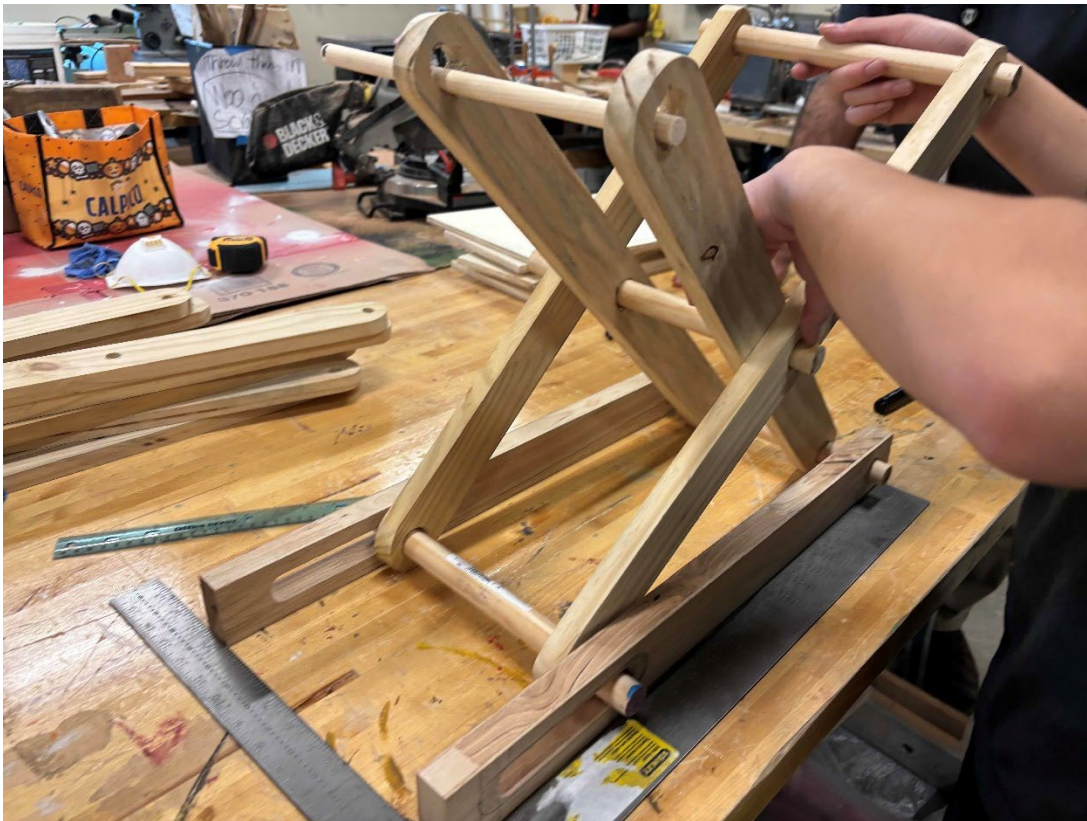


2/16/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

Now that we had our dowels, we were able to start cutting them and fitting them into the scissor arms. Each dowel was cut into 10" lengths to fit in the box. The bottom chassis of our robot is 10" inner width meaning we had to cut around that length. The bottom arms use $\frac{7}{8}$ " dowels while the other arms are connected through $\frac{1}{2}$ " dowels.

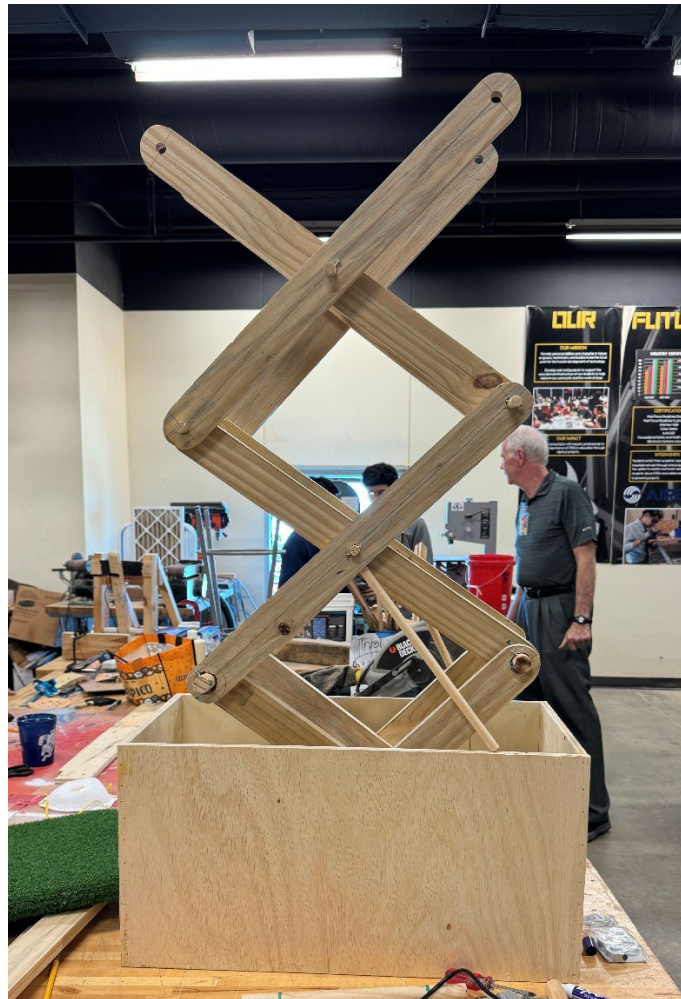


2/20/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: CREATE

Since we finished drilling the holes for our arms, all that is left is assembling the scissor lift, adding the wheels, and programming/wiring the motors. To assemble the scissor lift, we need to keep the arms in place without them sliding around on the rod connectors. To prevent this from happening, we decided to use screws that would go through the dowels and anchor the arms in place. Rather than drilling the screw through the arm and the dowel at the same time, we drill it through the dowel right next to the arm to keep it from moving in that direction. This is because the arms rotate, so screwing through it would prevent them from moving.



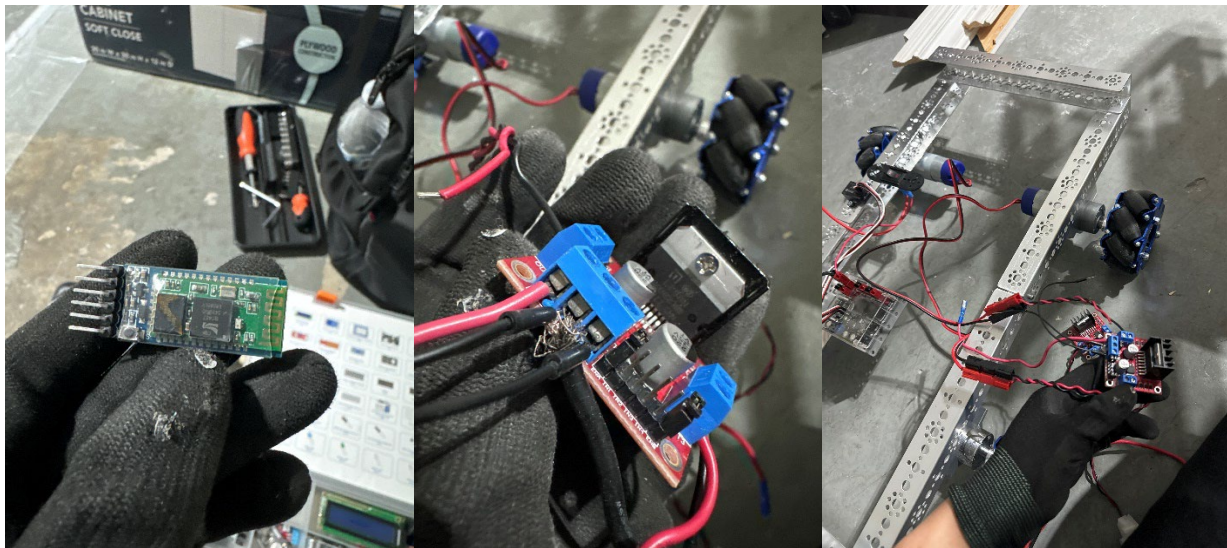
2/21/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: TEST

PROBLEM: Our Bluetooth module for the Arduino burnt out, meaning we will have to go back to using the Tetrax motor controllers with the RC controller for district competitions. This allows us to test the Tetrax modules, however. The Tetrax motor controller also only controls two motors at a time, meaning we can only control two wheels at once or one wheel and linear actuator.

SOLUTION: We will be using the Tetrax motor controllers for district competitions. Since this is only a prototype, this allows us to test that our actual mechanisms are working. When we present our prototype, we will first demonstrate the wheels then disconnect the wheels and switch it to the linear actuator.



2/22/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: IMPROVE

Today was our final day to make changes to our robot.

2/23/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: ASK

We finished district competition for our robot today. We asked the judges for feedback, and they recommended us to do market analysis and industrial reviews.

Now that we have more time, we will go back to using the Arduino module which will let us control all motors simultaneously.

2/26/2024

ENGINEERING JOURNAL

STAGE OF ENGINEERING PROCESS: ASK

PROBLEM: The Tetrix modules are not a suitable replacement for the Arduino. Our first prototype also tends to lean in directions and the wheels are not attached. The material may also need to be changed for our next design.

We are going to ask for feedback from several engineers and peers to see what else we can improve and what steps to take next.

Links to cite later or add to the research section:

<https://usafacts.org/articles/how-many-police-officers-die-in-the-line-of-duty/>

<https://www.news4jax.com/news/local/2021/09/24/traffic-stops-present-a-high-risk-to-law-enforcement/>

<https://www.theguardian.com/us-news/2022/apr/21/us-police-violence-traffic-stop-data>

<https://www.nbcnews.com/mach/science/police-robot-could-make-traffic-stops-safer-ncna1002501>

<https://www.nbcnews.com/news/us-news/police-search-suspect-after-california-officer-killed-during-traffic-stop-n952111>

<https://openpolicing.stanford.edu/data/>

<https://www.nhtsa.gov/data/traffic-records>

<https://bjs.ojp.gov/library/publications/contacts-between-police-and-public-2018-statistical-tables>

<https://www.ncsl.org/civil-and-criminal-justice/traffic-stop-data>

<https://data.austintexas.gov/Public-Safety/Racial-Profiling-Dataset-2015-Citations/sc6h-gr9f>

[Playlist of videos](#)

https://www.amazon.com/WindyNation-Actuator-Mounting-Brackets-Maintained/dp/B0787KGR5N/ref=sr_1_3?crid=EA97TOD6AVEI&keywords=windynation%2Blinear%2Bactuator%2B16%2Binch%2Bstroke&qid=1707249723&srefix=windynation%2Blinear%2Bactuator%2B16%2Binch%2Bstroke%2Caps%2C99&sr=8-3&th=1

https://www.amazon.com/gp/product/B00NM8H67C/ref=sw_img_1?smid=A1P855JR5KPOWZ&psc=1

<https://grabcad.com/library/linear-actuator-33>

